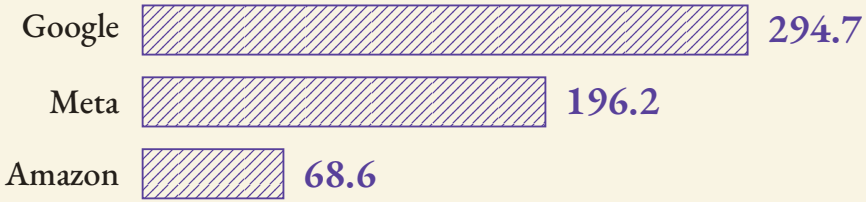
Three companies, 1.5 billion dollars of ads a day

Advertising revenue in 2025, in billions of US dollars

Alphabet and Meta: annual reports (Form 10-K) for 2025. Amazon: its results for the four quarters of 2025, added up.

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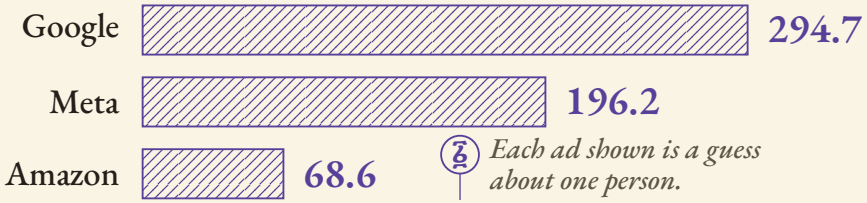
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Advertising revenue in 2025, in billions of US dollars



Together **559.5** billion, about **1.5** billion a day

Alphabet and Meta: annual reports (Form 10-K) for 2025. Amazon: its results for the four quarters of 2025, added up.

A machine suggested much of what we watch and buy

What share of your hours on YouTube did you go looking for?

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YouTube, watch time

Neal Mohan at CES 2018, in CNET

Share that came from a recommendation



over 70%

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Netflix, viewing

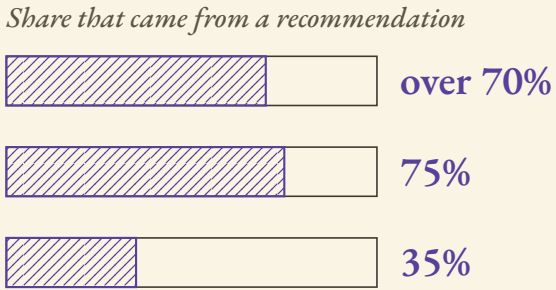
Netflix Tech Blog, 2012



75%

A machine suggested much of what we watch and buy

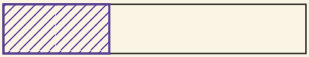
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The question to ask: compared to what?

Through a recommendation

Share of Amazon purchases that came from a recommendation



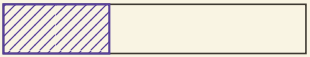
35%

McKinsey, 2013.

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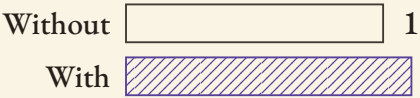


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McKinsey, 2013.

Caused by a recommendation

Randomized experiment: sales of a product and its recommended substitutes, on average



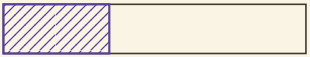
+11% more sales

Kumar and Hosanagar, Information Systems Research, 2019. Jannach and Jugovac, ACM TMIS, 2019.

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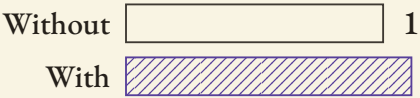
35%

A share also counts purchases that would have happened anyway.

McKinsey, 2013.

Caused by a recommendation

Randomized experiment: sales of a product and its recommended substitutes, on average



+11% more sales

Published field tests more often report 1% to 5% more revenue.

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What a better model added at Meta in 2026

More conversions after one change to a model, July to September 2026

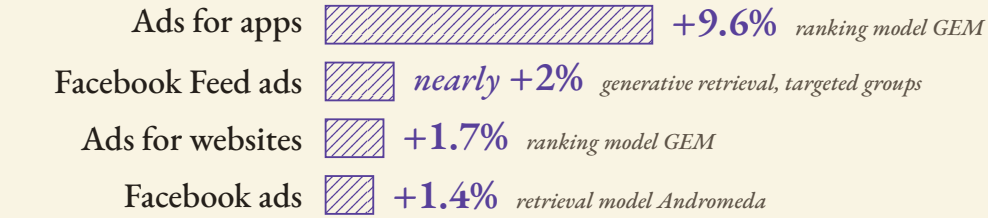
A conversion is what the advertiser wants from an ad: a purchase, an install, a sign-up.

Meta, earnings call of 21 October 2026: Susan Li, finance chief. No method given.

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Ads for apps	+9.6%	*ranking model GEM*
Facebook Feed ads	*nearly* +2%	*generative retrieval, targeted groups*
Ads for websites	+1.7%	*ranking model GEM*
Facebook ads	+1.4%	*retrieval model Andromeda*

Price per ad, in a year: +14%

Meta: driven by better ad performance and a better economy.

Meta, earnings call of 21 October 2026: Susan Li, finance chief. No method given.

400 million customers used a shopping assistant

Alexa for Shopping, Amazon

Users in the last 12 months

over 400 million

Interactions, change in a year

+570%

It compares products, checks a year of prices and suggests gifts.

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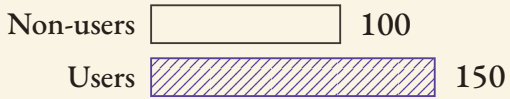
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Spending per order

US customers: users against non-users



+50% per order

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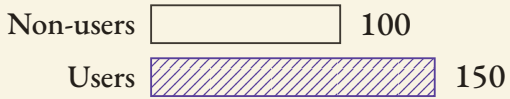
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Compared to what? The users chose the assistant themselves.

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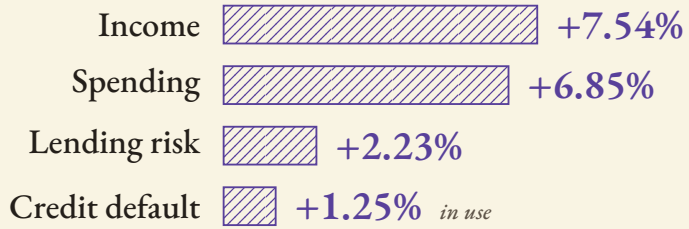
A profile learned from 100 billion transactions

Nubank: gain from adding the learned profile, each task on its own score

Braithwaite and others (Nubank), KDD 2026 Workshop on Machine Learning in Finance, arXiv 2507.23267.

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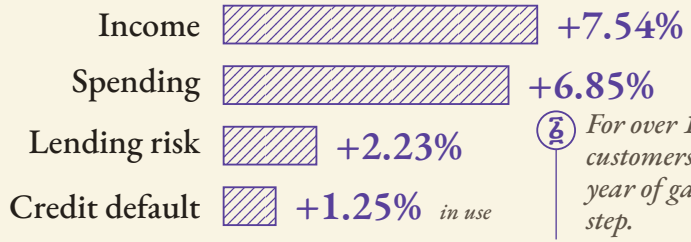
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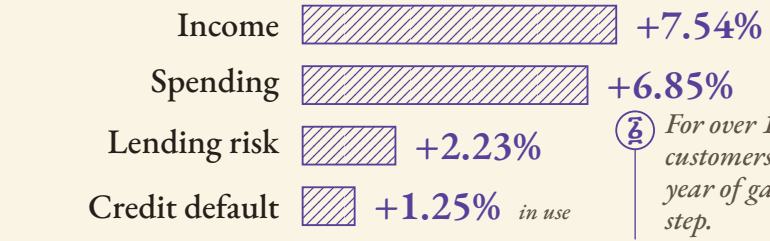
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Nubank: gain from adding the learned profile, each task on its own score



No new data: only a better reading of the bank's own transactions.

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